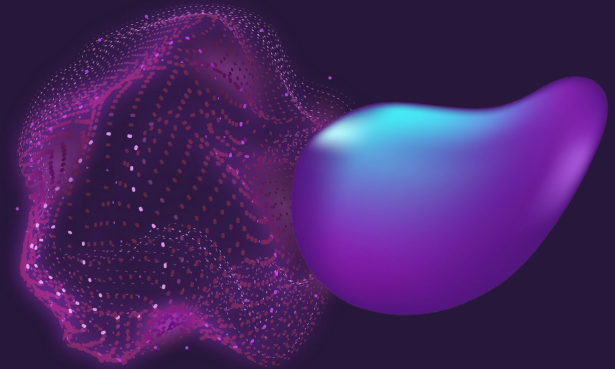




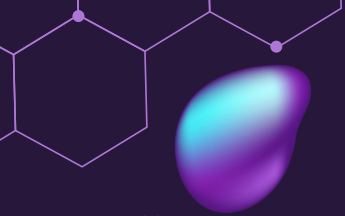
Pulse AI - ValveGuard

Dynamic Surveillance for
Bioprosthetic Valve Durability

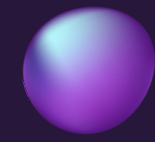
Athanasίου-Paraskevopoulou Maria – Medical Student
Chatzipapas Konstantinos – Electrical & Computer Engineering Student
Myaris Stylianos – Informatics Student

September 2026 – Dyania Health Hackathon





The Clinical Problem



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What is SVD and why durability matters:

Bioprosthetic valves wear out over time.

The shift: patients receiving are younger than ever & outlive their bioprostheses.

The SAVR vs TAVR Uncertainty

SAVR (Surgical): Durability documented over decades.

TAVR (Transcatheter): Unique procedural wear-and-tear, mechanical stress.

TAVR moves into younger patients **faster than the evidence.**

Current Workflow Failure

<u>Reactive Clinical Mindset</u>	Action triggered after symptoms - collapse
<u>Loss to Follow-Up</u>	Over 40% of post-AVR patients skip checkups once asymptomatic
<u>Measurement Variability</u>	Operator-dependent errors distort calculations
<u>Clinical Cost</u>	Irreversible myocardial remodeling, pulmonary edema, shock
<u>Financial Cost</u>	Emergent re-admissions, prolonged ICU stays, emergency procedures

The Surveillance Gap

Valve decay is silent until it is severe → standard annual follow-ups are too rigid.

Patients are often diagnosed 1 to 2 years too late – presenting to the ER with shortness of breath.

The stakes:

Caught early: <3% mortality

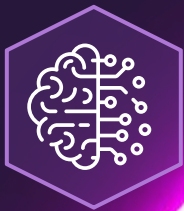
Caught late: - Emergency

Fix: >8-10% mortality



Our Hypothesis

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Learn

- from thousands of valves.
- Trained on serial echo trajectories across years of follow-up.

+



Reconstruct

- for this one valve.
- Builds the **individual patient's curve** of forecasted function from their own follow-up measurements, even predicting the time of failure – Structural Valve Deterioration (SVD).



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Anticipate

- before symptoms.
- The curve's shape is the information** – risk as a function of time.
 - flat = stable ·
 - sloped = ageing ·
 - accelerating = the warning.



Study Overview

Target Population

Adults with primary bioprosthetic SAVR or TAVR into native anatomy, with a baseline reference echo at 30–90 days and ≥ 1 later echo.

Exclusion

mechanical valve · valve-in-valve as index procedure
concomitant mitral/tricuspid replacement · active
endocarditis at the index operation · no baseline echo,

- *Event-free follow-up of patients who develop endocarditis, paravalvular leak/thrombosis, NSVD after implantation still contributes to the model.*

Endpoints

Primary: **the functional course of the valve and the transitions along it — normal $\rightarrow$ moderate $\rightarrow$ severe deterioration or failure-SVD (inc. SVD-related reintervention, SVD-related death)**

Secondary: all-cause reintervention · clinically expressive failure · paravalvular leak progression · non-structural failure · all-cause mortality = competing risk

Ground Truth Definition for SVD Y=1

Every criterion is change from the patient's own baseline echo. Adhering to VARC-3.

- Stenosis: gradient rise AND a concomitant fall in valve area or Doppler velocity index · Moderate $\Delta \geq 10 \rightarrow \geq 20$ mmHg · Severe $\Delta \geq 20 \rightarrow \geq 30$ mmHg
- Regurgitation: new or worsening intraprosthetic AR — Moderate ≥ 1 grade $\rightarrow \geq$ moderate · Severe ≥ 2 grades $\rightarrow$ severe
- A gradient that rises with no fall in area or velocity index is usually more flow, not a failing valve.

Confirmed, not single-shot: repeat echo ≥ 4 weeks later, or CT/TEE, or reintervention/death within 6 months

Evidence hierarchy: explant pathology > reintervention for a structural cause > serial echo..

All candidate events are adjudicated by an **independent clinical events committee**, blinded to model output — together with a random sample of records the algorithm called negative, so that missed events are detectable.

Why ValveGuard matters



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Ideal Next Follow-Up Date

Arbitrary & annual → **risk-adapted follow-up intervals.** (e.g. 3-6 months - high-risk vs 18-24 - stable low risk). Emergencies → scheduled re-do operations. Medical system relieved, **shifting slots toward patients who need them.**

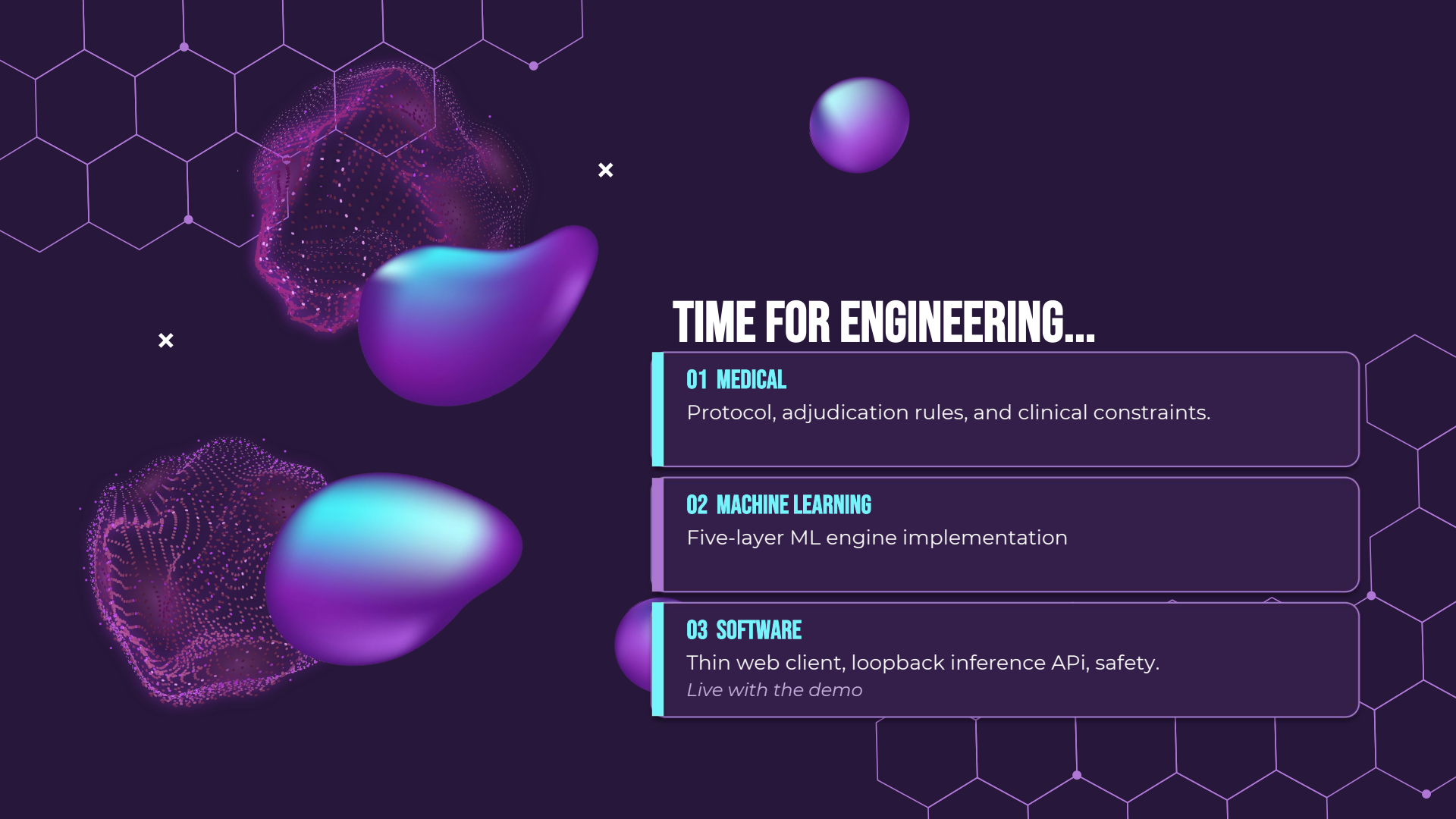
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Recognizing a Blind Spot

Deteriorating valves can sometimes hide behind normal gradients, as stroke volume falls. ValveGuard uses metrics such as **Doppler Velocity Index** and **Effective Orifice Area**, uncovering masked deterioration.

A Self-Auditing Model

Predicts what the next echocardiogram will show — **graded on it at the next visit.** Accumulated forecast accuracy becomes running prospective validation **at zero extra data cost.**



TIME FOR ENGINEERING...

01 MEDICAL

Protocol, adjudication rules, and clinical constraints.

02 MACHINE LEARNING

Five-layer ML engine implementation

03 SOFTWARE

Thin web client, loopback inference API, safety.

Live with the demo

FIVE-LAYER ENGINE

The medical team hands us the protocol. We compile it into five frozen layers: rules first, probability only when the case is eligible, action last and clinician-owned.

1

VARC-3 rules

2

**Competing-risk
CIF**

3

Echo trajectory

4

**Calibrate &
safety**

5

**Surveillance
interval**

RETURNED TO THE MEDICAL TEAM

Current HVD candidate · BVF / death cumulative incidence · next-echo path · calibrated risk or an explicit abstention · a review interval.

FIVE SUB-ENGINES

Input and output in clinical terms; mechanism underneath.

TIER 1

VARC-3 rules

In: reference vs current TTE

Out: no HVD / 2 / 3 / indeterminate

Pure function, no neural net

Candidate HVD, not SVD

Same case = same result

TIER 2

Competing-risk CIF

In: baseline + serial history

Out: BVF and death CIFs

Irregular-time GRU

Stage 2 stays on the BVF clock

NSVD class untrained — withheld

TIER 3

Echo trajectory

In: same history + state

Out: next gradient, Vmax, EOA, DVI, AR

Second GRU, two heads

Progression only after Stage 2/3

LVEF untrained — withheld

TIER 4

Calibrate & safety

In: raw CIF + missingness / OOD

Out: calibrated risk or abstain

PAVA isotonic maps, no extra net

Broken input → refuse to speak

Abstain is never shown as low risk

TIER 5

Surveillance policy

In: stage + 1-y risk + safety

Out: review now / 3 / 6 / standard

Clinician-owned step function

Draft policy → always review_now

Never chooses what to implant, provides tools for the medTeam

WHAT WE TRAINED ON

Hackathon proof of concept vs the protocol study the engine is written for.

THIS LAPTOP DEMO

Expanded synthetic notes — 1,500 linked patients, serial TTEs, 30–90-day reference.

De-identified real notes are inventoried and hashed, not used as labels (year-only dates).

REAL-WORLD DEPLOYMENT

2,500–3,000 patients · 10,000–15,000 patient-years · ≥150–200 adjudicated SVD-HVD events.

Multi-center registry, blinded committee, including a sample of algorithm-negative records.

HOW THE DEMO RUNS

THIN CLIENT

Static HTML/JS. No Node, no PyTorch in the browser, no checkpoints on disk in the UI.

LOOPBACK API

FastAPI on 127.0.0.1:8000. POST /v1/inference is the only runtime boundary.

HASHED ARTIFACTS

Tier 2/3 weights, calibration maps, and policy load only if SHA-256 matches the manifest.

CAUTION

Stage 3 or broken input withholds numeric risk.

SECURITY FOR THIS DEMO

LOOPBACK ONLY

The demo binds to 127.0.0.1. No public ingress. Stale listeners are killed before start-demo.

IF A REMOTE CLIENT IS NEEDED

Use a Tailscale overlay. Do not publish the inference API on the open internet.

FUTURE LLM USE

Keep PHI on a local model, or a contractually safe API. Never paste identifiable notes into a public frontier chat.

AUTHENTICATION

Today: smoke-test login for the hackathon UI. Any real record requires institutional SSO before data entry.

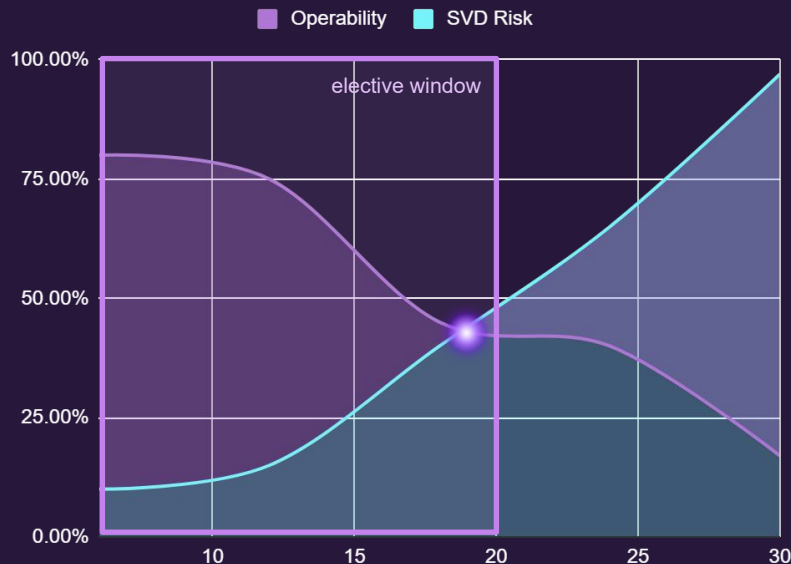
The Next Step – Exploratory Objective

- ◆ The Heart Team's real decision: *re*intervene *electively now*, while the patient is still a good candidate, or wait and risk operating *on a patient who has meanwhile become a worse one?*

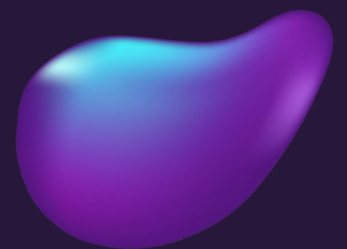
- ◆ Two curves, opposite directions — failure risk rises, **operability** falls ·

Would be utilizing data such as: Age, Renal function trend, LVEF and LV geometry · Comorbidity · Pulmonary pressures, RV function · Haemoglobin, albumin — frailty proxies · Prior heart-failure admissions · **EuroSCORE II / STS-PROM**

No validated model exists for the forward trajectory of operability at the time.



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**The clinically useful output is
where they cross.**



Thank you

PulseAI · ValveGuard — medical research to software

